

Place-Specific Reef Fish Identification for Coral Restoration

MARLYN ARQUE RUPA, University of California San Diego, USA

ALEKSA POPOVIC, University of California San Diego, USA

EWAN SHEN, University of California San Diego, USA

VIBUSHA VADIVEL, University of California San Diego, USA

KALEIGH EDUSADA, University of California San Diego, USA

This project develops an end-to-end pipeline for fish detection and species classification using underwater images provided by Coral Gardeners. We evaluated multiple detection and segmentation approaches, including YOLOv8, YOLOv11, YOLOv12, RF-DETR, and SAM3, and found that underwater imagery presents significant challenges due to domain shift, poor image quality, occlusion, and limited labeled data. Our final pipeline combines SAM3-based fish localization with BioCLIP-based species classification and incorporates region-specific species priors for Moorea, Fiji, and the Gulf of Thailand using marine biodiversity databases. Results show improved performance through stronger localization, region-aware classification, and higher-quality labels, providing a foundation for automated ecological monitoring in coral reef environments.

Additional Key Words and Phrases: reef fish identification, underwater computer vision, segmentation, classification, domain shift, coral restoration

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1 Introduction

Coral reef restoration projects depend on the ability to monitor reef ecosystems over time. Reef fish are especially useful for this purpose because their presence, abundance, and species composition can reflect changes in habitat quality, reef recovery, and ecological stress [6, 13]. A restored reef that supports diverse fish populations is more likely to function as a healthy ecosystem, whereas changes in fish communities may indicate degradation, pollution, habitat loss, or other environmental pressures. For organizations such as Coral Gardeners, image-based monitoring provides a practical way to collect large amounts of visual evidence from restoration sites. However, turning these images into insightful data remains difficult.

Traditional reef monitoring often relies on manual surveys, expert annotation, or visual inspection of underwater footage. These methods are accurate but do not scale well. They can be time-consuming, and are difficult to apply consistently across large image collections or multiple reef locations [5, 22]. Computer vision has the potential to reduce this bottleneck by automatically identifying fish, summarizing species distributions, and

Authors' Contact Information: Marlyn Arque Rupa, University of California San Diego, San Diego, California, USA; Aleksa Popovic, University of California San Diego, San Diego, California, USA; Ewan Shen, University of California San Diego, San Diego, California, USA; Vibusha Vadivel, University of California San Diego, San Diego, California, USA; Kaleigh Edusada, University of California San Diego, San Diego, California, USA.

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helping restoration teams compare reef activity across sites and time periods [5]. However, reef imagery is a challenging setting for modern vision models. Underwater images often contain low contrast, motion blur, poor lighting, color distortion, partial occlusion, background clutter, and small distant fish [20, 21]. In addition, models trained on cleaner public datasets often do not generalize well to field imagery collected under different environmental conditions. This domain shift was one of the main technical challenges encountered during the project [7, 19].

This project develops a place-specific reef fish identification pipeline for underwater coral reef imagery. The goal is to build a practical end-to-end system that can process raw underwater images, locate fish-like regions, classify detected crops, and produce lightweight metadata that could support ecological monitoring workflows. The system follows a two-stage design. First, a segmentation model localizes fish in the image and generates candidate regions. Second, a biological vision-language classifier predicts the most likely species for each accepted crop. This separation is important because localization and classification fail in different ways. A model may correctly recognize a fish species from a clean crop but fail when the fish is small or partially hidden in the full reef image. Similarly, a detector may locate a fish-shaped object but still provide a crop that is too blurry, incomplete, or visually ambiguous for species classification.

At the beginning of the quarter, the project was framed as a segmentation-and-classification pipeline that would be trained, improved, and evaluated on location-specific reef data. As the project progressed, the work evolved into a broader experimental study of underwater model reliability. Early experiments showed that models trained on public fish imagery did not automatically transfer to Coral Gardeners footage. In particular, custom YOLO-based detectors [10] performed poorly under real reef conditions, while RF-DETR [11] improved localization but produced many false positives, especially on buoy-like objects. These results shifted the project from simply tuning one model to understanding the relationship between model choice, data quality, annotation completeness, and domain shift.

The final system uses SAM3 [2] for fish localization and BioCLIP 2.5 [16] for species classification. SAM3 was selected because it produced stronger localization results and cleaner crop regions than the earlier detection approaches tested in the project. BioCLIP 2.5 was selected because it provides a biology-oriented image-text classification framework that can be constrained using candidate species names. To make classification more ecologically realistic, the pipeline uses region-specific species priors for Moorea, Fiji, and the Gulf of Thailand derived from open marine biodiversity occurrence data [4]. These priors restrict the classifier to species that are likely to occur near the reef location being analyzed, reducing the chance that the model predicts visually similar species from unrelated regions. The system also supports optional preprocessing and debugging outputs, including tiling, contrast enhancement, saved crops, masks, overlays, and metadata summaries.

The main contributions of this project are:

- We built a reproducible end-to-end reef fish identification pipeline that accepts underwater images or image folders, localizes fish-like regions, generates crops, classifies species, and exports structured CSV/JSON metadata.
- We evaluated multiple localization approaches, including YOLO, RF-DETR, and SAM3, and showed that underwater domain shift strongly affects model performance on Coral Gardeners imagery.
- We incorporated region-aware classification by using top-25 species priors for Moorea, Fiji, and the Gulf of Thailand, allowing BioCLIP predictions to be constrained by ecological location.
- We tested preprocessing and tiling strategies for improving segmentation performance, showing that tiling improved recall and F1 score but increased runtime, making it useful as an optional high-recall mode.
- We conducted failure-case analysis and identified buoys and other non-fish objects as major sources of false positives, motivating the creation of hard-negative annotations for future retraining.

- We organized the implementation into a reusable package with documented scripts, configuration files, evaluation tools, and optional visualization artifacts to support future development beyond the course project.

Overall, this report presents the technical design, experiments, and lessons learned from building a computer vision pipeline for reef fish identification. The project demonstrates that automated reef fish monitoring is feasible, but also shows that performance depends heavily on dataset quality, annotation consistency, and the match between public training data and real deployment imagery. The rest of the report describes related work in underwater fish detection and segmentation, the structure of the pipeline, the datasets and region priors used, the segmentation and classification experiments, the main failure cases, and the milestones completed over the quarter.

2 Related Work

2.1 Fish Detection

There has been previous work where deep learning was used for underwater fish detection and classification, especially because identifying fish underwater from reef images is very time consuming and can be difficult. Zhao et al. [22] focused on computer vision models to detect and classify fish species from underwater images. Specifically, they use Composited FishNet, which is a convolutional neural network-based detection algorithm, to handle low resolution, low image quality, poor lighting, and fish camouflage. Additionally, Gregory et al. [5] developed a real-time fish detection system using a MobileNet SSD model. They integrated edge computing to detect fish from videos.

FishNet and MobileNet are related to our project because they demonstrate how computer vision models can be used for detecting fish in underwater environments. They also encounter similar underwater image challenges that we faced, such as poor lighting, low resolution, low image quality, and camouflage.

2.2 Segmentation Models for Underwater Images

Segmentation is a very important step for underwater fish identification. This is because the model needs to locate the fish before it can classify the species reliably. Xu et al. [20] introduced AquaSAM, where they focused on improving the Segment Anything Model (SAM) for underwater fish segmentation, with the goal of outperforming the default SAM. Just like in our project, they found that segmentation models are not always the best at segmenting in underwater conditions and environments. Yang et al. [21] introduced WaterSAM, where they enhanced the image encoder to lower the amount of trainable parameters and reduce computational costs.

AquaSAM and WaterSAM show the importance of segmentation quality when working with underwater footage, especially when images are blurry or fish are partially covered by coral. This connects to our project because segmentation is an important step in our pipeline, so an unreliable segmentation model could affect the accuracy of the entire system.

2.3 Domain Shift Between Public Datasets and Real Reef Footage

Domain shift is a huge challenge when working with reef images, especially when working with computer vision models. This is because a model can perform well on a clean and clear dataset, but then may not generalize well when used in a new environment, such as real reef imagery. Wen et al. [19] introduced EnYOLO, which addresses domain shift by combining image enhancement with object detection so that the model can adapt better to underwater environments. Liu et al. [7] proposed Water Quality Transfer (WQT) with DG-YOLO to improve domain generalization. They used WQT to increase diversity in the training data, which helped improve fish detection when encountering domain shifts.

Since domain shift was a key issue throughout our project, these papers highlight similar issues with underwater imagery and inconsistent datasets. In our project, we dealt with domain shift between public datasets such as iNaturalist and the images provided by Coral Gardeners.

2.4 Coral Reef Datasets

A major challenge for reef fish detection is the lack of large, high-quality, and well-labeled datasets. When training our models, well-labeled and annotated images are needed, but labeling and annotating by hand is very time consuming. Wang et al. [18] introduced SCSFish2025, a dataset from the South China Sea with high-resolution images, many unique species, and detailed labels. Their goal is to address this challenge and support automatic detection and identification for computer vision systems. Sauder et al. [15] introduced Coralscapes, another dataset that focuses more on semantic scene images from coral reefs while also providing segmentation labels for multiple benthic classes. Since Coralscapes also segments classes like coral, sand, and rocks, it helps models better understand the entire reef environment rather than only focusing on fish.

These reef datasets provide more labeled images of coral reef environments, which can be beneficial for training computer vision models. During our project, we had limited and inconsistent labels and data, which made training and evaluation more difficult. The Coral Gardeners images also had complex reef backgrounds, making it harder for SAM3 to separate fish from the surrounding environment.

2.5 Biology Focused Classification

The classification of species is an important aspect of our project. Nakkab et al. [9] used LiT to be able to efficiently classify species. They adapted computer vision models by using the iNaturalist dataset so they could utilize pretrained models to better classify species, especially when species could be very similar and sometimes training would not be enough to properly classify them apart. Another biology-focused classification model is TaxaBind. Sastry et al. [14] introduced TaxaBind as a shared embedding space for ecological applications. It does not only use images for classification, since it also uses information like text, location, audio, and environmental features.

Since we used BioCLIP in our project, which is a biology focused classification, LiT and TaxaBind are other biology-focused classification models. Both works show the importance of using biological labels and ecological context for species classification. This is important for our project because our goal is not only to detect fish, but also to classify them into species that are likely to appear in specific reef regions.

Related Works Summary. Overall, all of these related works show that reef fish segmentation and classification are all connected together as they are all important (detection, segmentation, quality of the data being used, domain shifts and adaptation techniques, and classification). These papers highlights why working with underwater imagery is challenging, since the models that are being used must be able to handle poor lighting, low resolution, limited labeled datasets, and reef environments. Thus, our project was able to built on each of these ideas as we need good data and good computer vision models for segmentation and classification for our images of reef environments.

3 Technical Material

3.1 System Overview

The final system follows a two-stage pipeline for reef fish identification, containing fish segmentation and species classification. Our team chose the two-stage pipeline design because localization and classification present different challenges in underwater imagery. Detection models must identify fish under conditions such as low contrast, motion blur, clutter, and murky reef backgrounds. Classification models must distinguish fish among

visually similar species in terms of color, shape, and size. By decoupling these tasks, our team was able to independently improve either stage, and errors can be isolated and analyzed more effectively.

Our pipeline begins with a reef image or image folder as input. Optional preprocessing techniques, including image tiling and contrast enhancement, may be applied to improve detection performance on small or low-visibility fish at the cost of runtime. The processed image is then passed to SAM3, a foundation object segmentation model [2], which generates fish candidate masks and bounding regions. Post-processing filters detections by applying Non-Maximum Suppression (NMS) to remove duplicate or overlapping predictions and bounding boxes.

The filtered detections are converted into padded image crops extracted from the original image and are passed to the BioCLIP 2.5 biology classification model [16] for species classification. Classification is constrained using a region-specific species prior of the top 25 frequently observed species, rather than allowing predictions from the full BioCLIP label space. Candidate species lists were generated for Moorea, Fiji, and the Gulf of Thailand using marine biodiversity observations from OBIS (Ocean Biodiversity Information System) [4], an open-access database. Species lists can also be automatically generated and supplied to the classifications stage, given a longitude, latitude, and radius user input. The pipeline restricts predictions to species known to occur within a selected region to improve ecological relevance and reduce confusion among visually similar species.

The pipeline produces lightweight outputs by default, including detection metadata, species predictions, and summary statistics stored in CSV and JSON formats. Additionally, optional debugging artifacts such as segmentation masks, overlays, and saved crops can also be exported to support qualitative model evaluation and analysis. Figure 1 visualizes the pipeline architecture and data flow.

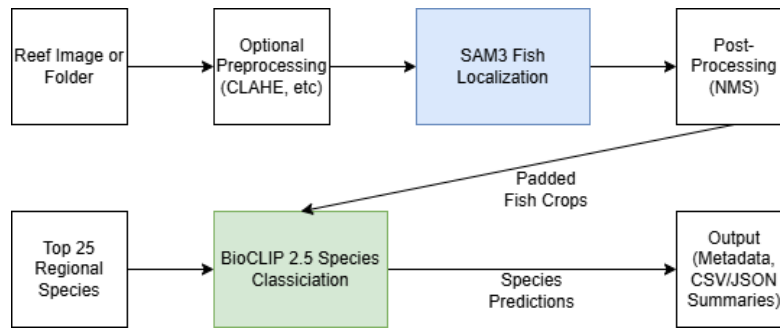


Fig. 1. Detection and Classification Pipeline Overview.

3.2 Datasets and Region Priors

The project utilized multiple datasets to support model development, evaluation, and validation. Coral Gardeners provided the primary underwater reef imagery dataset that we used to evaluate our models. The dataset images represented realistic deployment conditions for Coral Gardeners but introduced significant challenges, including low image quality, incomplete annotations, color distortion, partial occlusion, and murky reef backgrounds. The Coral Gardeners dataset represented the most important and difficult dataset we used throughout the project.

To evaluate model behavior under more controlled conditions, we incorporated additional public datasets into the development workflow. The DeepFish dataset [12] was used as a segmentation benchmark to assess localization performance across a controlled underwater domain. For species classification experiments, GBIF [3] and iNaturalist [17] image datasets were used to provide cleaner examples of reef fish species and enable region-specific evaluation of BioCLIP.

Furthermore, we incorporated region-specific species priors to restrict candidate species based on the geographic location of the reef being analyzed, rather than allowing the classifier to predict from the full set of fish species known to BioCLIP. Based on directions from Coral Gardeners, species lists were generated for Moorea, Fiji, and the Gulf of Thailand using occurrence data from the mentioned OBIS marine biodiversity database [4]. For each region, observations were queried within a geographic radius, ranked by occurrence frequency, and our system kept the 25 most frequently observed species.

The resulting species lists were stored in a YAML file and supplied to BioCLIP 2.5 during inference. We constrained predictions to species known to occur within the selected region so that the classifier model incorporated ecological context into the decision-making process and reduced confusion between visually similar species that are unlikely to geographically coexist.

Regarding dataset sizes, the primary Coral Gardeners YOLO-format dataset contained 1,778 reef images from the Moorea region with 6,893 labeled fish bounding boxes. Additional evaluation subsets were used for targeted experiments, including a 274-image SAM3 evaluation subset and a 25-image preprocessing ablation subset. Classification experiments used Coral Gardeners' crops/predictions and a 107-sample iNaturalist-style comparison set.

3.3 Segmentation Model Evaluation

Fish segmentation was the most critical stage of the system pipeline due to the classification performance's dependence on the quality of the extracted fish crops. A detection was considered a true positive when its intersection-over-union (IoU) with a ground-truth annotation exceeded 0.5. The main Coral Gardeners segmentation evaluation was run on the 1,778-image dataset. Precision, recall, and F1 were computed by matching predicted boxes to labeled fish boxes using this IoU threshold. Early experimentation focused on evaluating multiple detection and segmentation approaches, including custom-trained YOLO models, RF-DETR, and SAM3. Our team conducted the segmentation experiments and evaluation to determine which approach performed the best on representative reef imagery.

3.3.1 YOLO Experiments. The first detection experiments used custom-trained YOLOv8, YOLOv11, and YOLOv12 models trained on approximately 7,000 labeled fish bounding boxes from publicly available iNaturalist data. YOLO (You Only Look Once) [10] is a deep learning algorithm for object detection. While our trained YOLO models performed well on a validation set from the iNaturalist dataset, they generalized poorly to the Coral Gardeners reef images. The models struggled with differences in lighting conditions, water clarity, fish appearance, and image quality between the training and deployment environments, representing the heavy domain shift challenge.

Quantitatively, the YOLO models only achieved 6.9% precision and a 7.9% F1 score on the Coral Gardeners evaluation dataset, as seen in Table 1 below. Manual visual inspection revealed difficulty identifying fish in low-resolution underwater images and missed segmentation of partially hidden fish. Due to these limitations, our team did not pursue the YOLO-based approach any further for the final pipeline.

3.3.2 RF-DETR Evaluation. To improve upon YOLO's shortcomings and localization performance, the project next explored RF-DETR, a transformer-based object detection model [11]. RF-DETR greatly outperformed the YOLO baselines, achieving 49.6% precision, 41.5% recall, and a 45.2% F1 score. We inferred that transformer-based detectors were more robust to real-world underwater environments compared to our trained YOLO models.

However, the RF-DETR approach had a significant false-positive issue. After manual visual analysis, we discovered that the model frequently detected non-fish objects as fish, especially buoys. Particularly for the Coral Gardeners evaluation dataset, RF-DETR recorded more than 3,000 false-positive detections, significantly hindering segmentation performance. Overall, while RF-DETR improved fish localization, the false-positive issue reduced the suitability for later species classification.

3.3.3 SAM3 Segmentation. For our final segmentation approach, we used SAM3, a foundation segmentation model that generates object masks across diverse visual domains with a given user prompt [2]. Unlike traditional object detectors such as YOLO or RF-DETR, SAM3 produces segmentation masks that more comprehensively capture fish boundaries and improve crop quality for later species classification.

Among all evaluated segmentation approaches, SAM3 achieved the highest F1 score with tiling preprocessing discussed in Section 3.4, producing 77.8% precision, 68.1% recall, and a 72.9% F1 score on the Coral Gardeners dataset, as seen in Table 1. Figure 2 below visualizes the comparison of the fish detection model performances. In addition to improved quantitative performance, SAM3 produced cleaner fish crops and fewer localization failures than the previous approaches. Based on both quantitative and qualitative evaluations, our team selected SAM3 for the segmentation stage of the final pipeline.

Table 1. Comparison of Fish Detection Models on the Coral Gardeners evaluation set.

Model	Precision (%)	Recall (%)	F1 Score (%)
YOLOv8/v11/v12	6.9	–	7.9
RF-DETR	49.6	41.5	45.2
SAM3	77.8	68.1	72.9

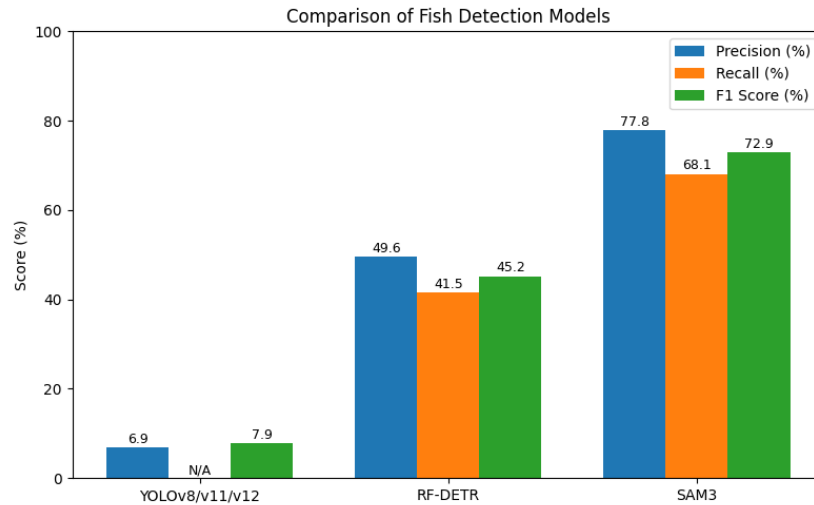


Fig. 2. Comparison of Fish Detection Models for Precision, Recall, and F1 Score.

3.4 Image Preprocessing Experiments

To further improve fish segmentation performance under underwater reef imagery conditions such as blurry contrast, poor lighting, and color distortion, we evaluated several preprocessing techniques using the SAM3-based detection pipeline. Our goal was to determine whether image preprocessing techniques could improve segmentation performance with reasonable runtimes.

We investigated several preprocessing techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE), tiling, gray-world color correction, and gamma correction. CLAHE potentially distinguishes fish

boundaries by improving contrast by distributing pixel intensity distributions [8]. Tiling splits the input image into "tiles" to reduce the number of pixels processed at a time and improve detection of smaller objects. The method increases the relative size of the fish in the processed area. Gray-world correction assumes a neutral color for the average scene color, and attempts to shift potentially noisy pixels closer to the target neutral color [1]. Gamma correction adjusts image brightness through intensity transformations to shift the image closer to how the human eye perceives light. The technique applies a mathematical nonlinear curve to shift image luminance values from a linear scale to a logarithmic scale, closer to how the human eye perceives light.

To facilitate experimentation runtime, we tested each ablation of a 25-image sample from the Coral Gardener dataset. The experiment results can be seen in Table 2. The baseline method with no preprocessing achieved a precision of 83.5%, a recall of 56.6%, and an F1 score of 67.5%. For the rest of the preprocessing methods, CLAHE achieved the highest precision, tiling greatly raised recall, and had the largest F1 score, gray-world slightly lowered precision while raising recall, and gamma correction had no measurable impact on performance. The results express that improving spatial coverage offered more lift than color enhancements, likely due to the performance bottleneck being the small distant fish that are missed by the detection models.

As mentioned in Table 2, tiling increased recall from 56.6% to 68.1% and improved the F1 score from 67.5% to 72.9%. Combining tiling with Gray-world correction further increased recall to 69.7%. The SAM3 result in Table 1 corresponds to the tiled SAM3 configuration, which achieved the highest F1 score. Although the no-tiling baseline in Table 2 had higher precision, tiling preprocessing improved recall enough to produce the best overall F1 score. Regarding runtime, tiling increased overall runtime per image from approximately one second per image to three seconds per image, as the segmentation model had tiles to individually process. While tiling improved recall and F1, the approximate tripling of the runtime per image may not be acceptable for all deployments. As a result, we kept tiling as an optional high-recall mode rather than a default step in the pipeline. The preprocessing experiments emphasize that image preparation can improve segmentation performance, and the largest gains were obtained through improved spatial coverage rather than pixel color alterations.

Table 2. Preprocessing and Tiling Ablation Results on a 25 Image Sample from the Coral Gardeners Dataset

Method	Precision	Recall	F1	Time/Image
Baseline	0.8353	0.5657	0.6746	~1 s
Tiling	0.7844	0.6813	0.7292	~3 s
CLAHE Only	0.8466	0.5498	0.6667	~1 s
Gray-world Only	0.8197	0.5976	0.6912	~1 s
Gamma Only	0.8353	0.5657	0.6746	~1 s
Tiling + CLAHE	0.7885	0.6534	0.7146	~3 s
Tiling + Gray-world	0.7642	0.6972	0.7292	~3 s

3.5 Region-Aware Classification with BioCLIP

After fish segmentation, each detected fish crop was passed to a species classification stage. Our team experimented with two biology classification models as discussed later in the section. Due to BioCLIP 2.5's initial and consistent success and recommendation from Coral Gardeners, we chose BioCLIP 2.5 for the classification step of our final pipeline. BioCLIP is a biodiversity-focused vision-language foundation model. BioCLIP uses text encoders to map images to an embedding space with established biological species, families, genus, and more. Organisms within an image are encoded within the space and labeled with the closest established species embedding [16].

One major challenge in fish classification is the similarity of many species in terms of color, shape, and size. As a result, our team incorporated region-specific species information into the classification process. Regions of

interest included Moorea, Fiji, and the Gulf of Thailand, and a list of the 25 most commonly observed fish species was generated using frequency data fetched from the OBIS ocean biodiversity database [4]. BioCLIP predictions were restricted to the species in the incorporated frequency *yaml* file during inference. Our approach reduced the classification search space and mitigated the issue of the classification model being confused by visually similar fish species across the world.

We experimented with two BioCLIP models, including BioCLIP 2 and BioCLIP 2.5 with regional species information. BioCLIP 2 achieved 49.66% accuracy on the Coral Gardeners dataset, and BioCLIP 2.5, combined with region-specific species filtering, achieved 83.33% accuracy, as seen in Table 3. Figure 3 serves as a visualization of our classification results. The BioCLIP classification evaluation included 412 cropped images from the Coral Gardeners dataset. Additionally, classification accuracy is reported as top-1 accuracy on matched fish crops. Our experiment changes both the BioCLIP model version and the candidate species list, so the experiment should be treated as a practical comparison rather than an isolated ablation test. Additionally, while we did not retrain either model, and cannot attribute the lift to an architectural change within the two versions, our results emphasize that incorporating regional species information can improve species identification by reducing classification model confusion between visually similar fish species not expected within the same ecosystem.

Table 3. BioCLIP Classification Results on Coral Gardeners Dataset

Method	Accuracy (%)
BioCLIP 2	49.66
BioCLIP 2.5 + Region Priors	83.33

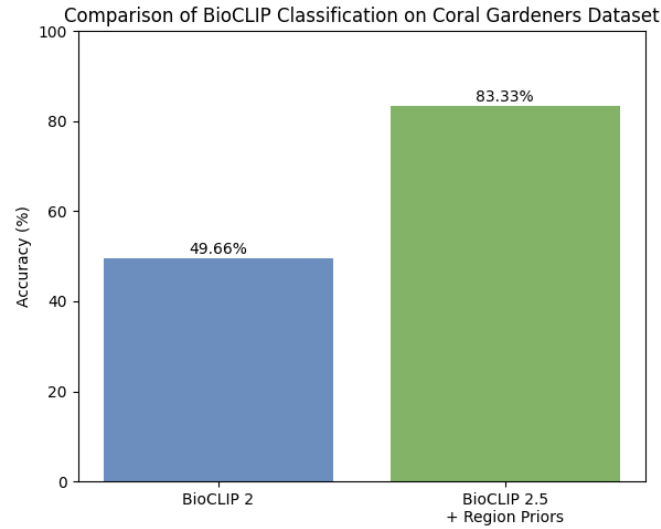


Fig. 3. Comparison of BioCLIP Classification on Coral Gardeners Dataset.

Finally, the classification stage outputs a species prediction for each accepted fish crop along with the relevant metadata. Fish species predictions are aggregated and stored in CSV and JSON formats, allowing Coral Gardeners to efficiently analyze species distributions and biodiversity statistics. The pipeline's lightweight output data avoids excessive storage overhead and provides a practical interface for Coral Gardeners' ecological reef monitoring.

3.6 Failure-Case Analysis and Hard Negatives

One of the most important findings during the quarter was that buoys were a major source of false positives. Because buoys share some visual properties with fish in underwater scenes, they were repeatedly detected as fish-like objects. To address this, the team began building an annotated buoy dataset and corresponding COCO-style artifacts so that buoys can be used as explicit hard negatives during retraining. However, the hand-labeled annotations were completed too late in the project timeline to support retraining, validation, and evaluation before deadlines.

This is a meaningful technical improvement because it attacks the dominant failure mode directly. Instead of only tuning model thresholds or retraining on more fish data, the approach emphasizes that data quality is vital to model performance.

4 Milestones

At the beginning of the quarter, the planned project was to build a two-stage fish detection and classification pipeline using a segmentation model followed by a YOLO classifier that we would need to train, fine-tune, test, and evaluate on a location-specific dataset. The original milestones were to establish a baseline model, improve robustness through pre-processing and augmentation, have a complete CLI-based inference system, optimize model performance, and document.

As the quarter progressed, we identified several challenges that were not anticipated in the original plan. The Moorea dataset contained many visible fish that were not accurately annotated, making the evaluation unreliable, and a significant domain shift was required to match the underwater reef imagery conditions for testing. For these reasons, the project shifted from improving a single model to experimenting with and comparing multiple segmentation and classification models and understanding how much data quality affects overall performance.

Our team quickly adapted its milestones to better address these issues while achieving the project's high-level goals. The following milestones were completed:

- End-to-end pipeline capable of detecting fish and generating crops for classification.
- Implementing a region-specific classification framework that narrows the number of fish tags based on geographic location.
- Evaluations of multiple segmentation and classification models were compared using several datasets instead of solely relying on a single model or architecture.
- Failure-case analysis focused on buoys, false positives.
- Establishing reproducible workflows for dataset preparations, preprocessing, and model training.

The milestone that evolved along with our work was model optimization. Initially, the goal was to improve performance through robustness techniques. However, our experiments showed that our bottleneck was data quality, with misleading annotations and domain adaptation having a greater impact on performance than model tuning. Our team focused on data analysis and evaluation to develop a more reliable understanding of the system's strengths and limitations, rather than continuing to tune a single model.

We managed these challenges by refocusing our efforts, running multiple experiments, using additional datasets for comparison, and leveraging location to find targeted datasets.

Overall, although the implementation strategy differed from the proposed plan, the project's major objectives were achieved. Our team completed an end-to-end pipeline that detects fish and generates crops for classification. We analyzed common failure cases, such as buoys and other non-fish objects, being detected.

5 Conclusion

This project developed and evaluated a place-specific reef fish identification pipeline for coral restoration workflows. The work demonstrates that an end-to-end fish detection and classification system is feasible, but that

underwater domain shift, incomplete annotations, and visually confusing non-fish objects are major obstacles to high accuracy. The team addressed these issues through model comparison, dataset engineering, BioCLIP integration, and hard-negative buoy annotation.

The broader significance of the project is that it provides a practical foundation for reef monitoring systems that are tailored to specific locations rather than treated as generic image classifiers. Future work should focus on stronger preprocessing, improved annotations, more robust segmentation training, and a cleaner user-facing interface for deployment and evaluation.

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Ethics and Privacy Statement

This project uses ecological imagery for conservation-oriented analysis and does not involve human subject data. The primary societal risk is that imperfect models could produce misleading biodiversity summaries if deployed without careful validation, especially under domain shift or incomplete annotation conditions. To reduce misuse, the project emphasizes transparent evaluation, clear documentation of failure modes, and reproducible outputs rather than unsupported automated claims.

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